

aula_pratica_agrupamento_com_dendrograma

June 16, 2026

```
[16]: # Agrupamento hierárquico com dendrograma

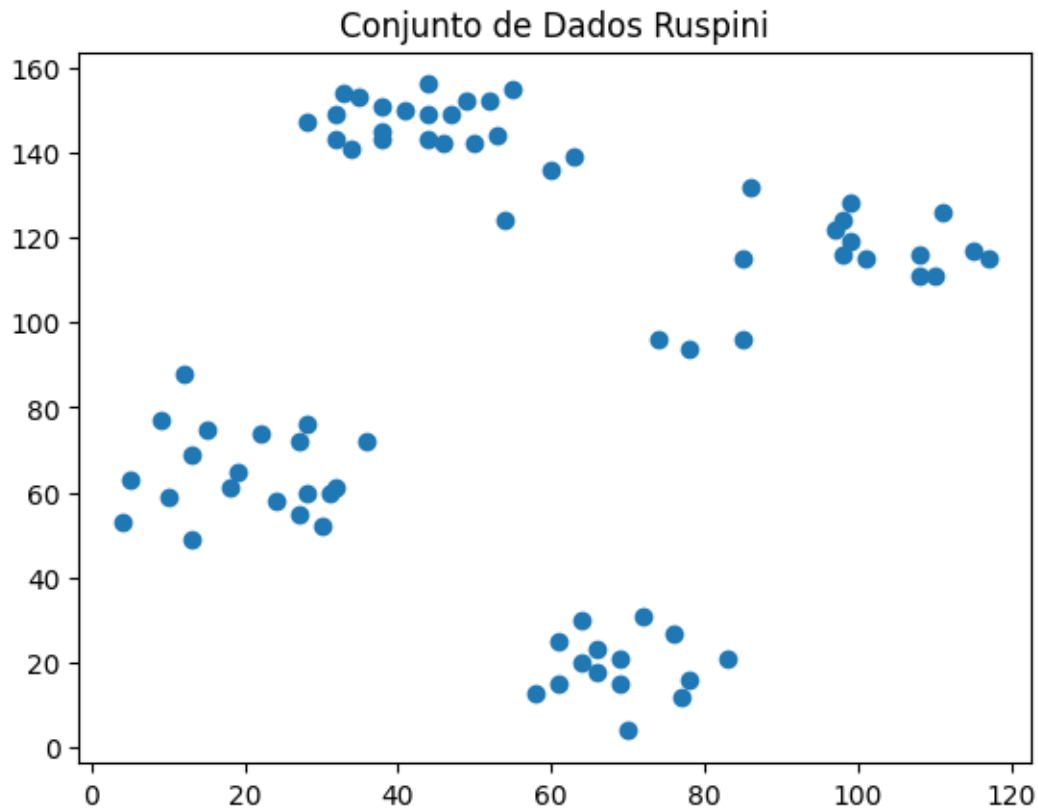
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import make_blobs
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import AgglomerativeClustering
from scipy.cluster.hierarchy import dendrogram, linkage
import pandas as pd
```

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[17]: df = pd.read_csv('ruspini.csv')

X = df[['x', 'y']].values

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

plt.scatter(X[:, 0], X[:, 1])
plt.title("Conjunto de Dados Ruspini")
plt.show()
```



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[18]: # 3. Gerando a matriz de ligação
# Métodos comuns: 'ward', 'complete', 'average', 'single'
linked = linkage(X_scaled, method='ward')
linked
```

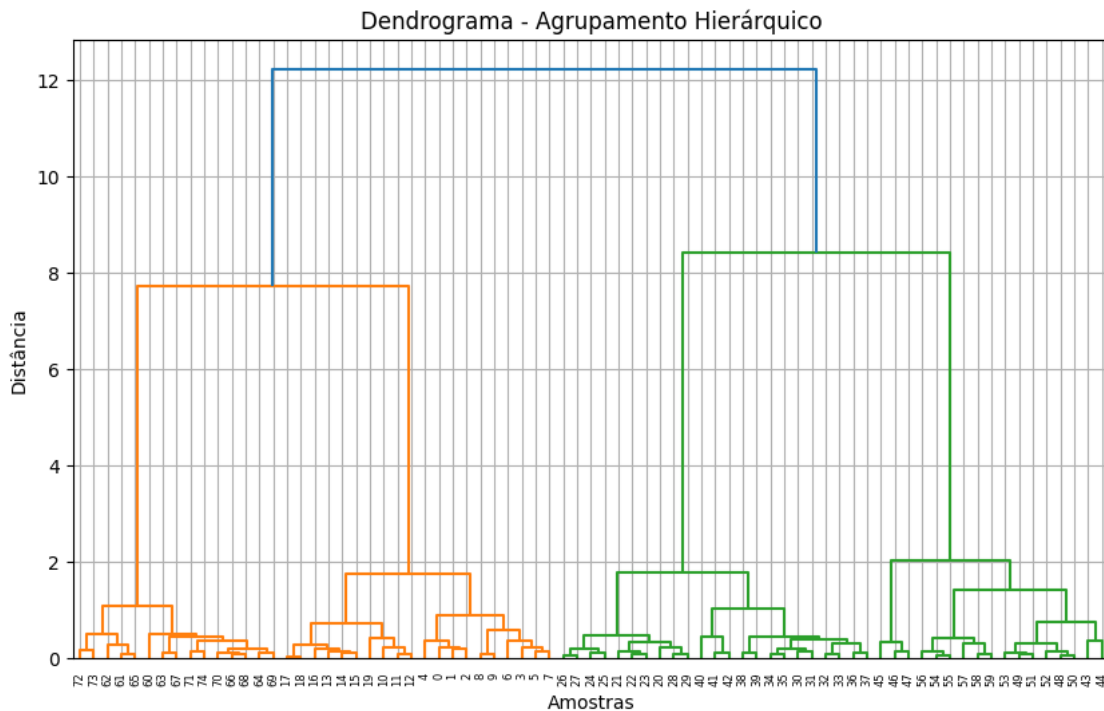
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[18]: array([[1.70000000e+01, 1.80000000e+01, 3.89437801e-02, 2.00000000e+00],
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 [7.00000000e+01, 8.30000000e+01, 1.02800777e-01, 3.00000000e+00],
```

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 [3.80000000e+01, 3.90000000e+01, 1.16831340e-01, 2.00000000e+00],
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[1.18000000e+02, 1.23000000e+02, 7.59493815e-01, 8.00000000e+00],
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[1.46000000e+02, 1.47000000e+02, 1.22104019e+01, 7.50000000e+01]]])
```

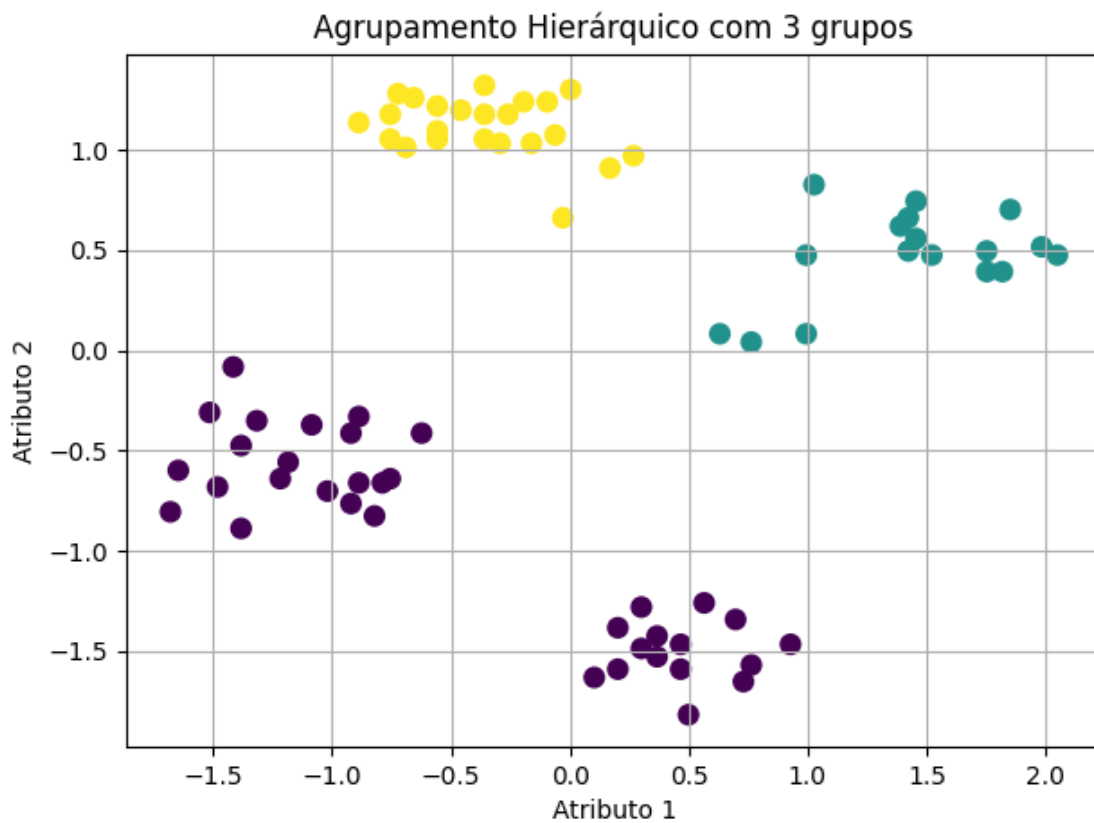
```
[19]: # 4. Plotando o dendrograma
plt.figure(figsize=(10, 6))
dendrogram(linked)

plt.title("Dendrograma - Agrupamento Hierárquico")
plt.xlabel("Amostras")
plt.ylabel("Distância")
plt.grid(True)
plt.show()
```



```
[20]: # Aplicando agrupamento hierárquico com 3 grupos
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```
modelo = AgglomerativeClustering(  
    n_clusters=3,  
    linkage='ward'  
)  
  
labels = modelo.fit_predict(X_scaled)  
  
plt.figure(figsize=(7, 5))  
plt.scatter(X_scaled[:, 0], X_scaled[:, 1], c=labels, s=60)  
  
plt.title("Agrupamento Hierárquico com 3 grupos")  
plt.xlabel("Atributo 1")  
plt.ylabel("Atributo 2")  
plt.grid(True)  
plt.show()
```



```
[21]: plt.figure(figsize=(10, 6))  
dendrogram(linked)
```

```
# Linha horizontal simulando um corte no dendrograma
plt.axhline(y=4, linestyle='--')

plt.title("Corte no Dendrograma")
plt.xlabel("Amostras")
plt.ylabel("Distância")
plt.grid(True)
plt.show()
```

